DATA CLEANING DOCUMENTATION

AI-Powered Predictive Bed Demand Forecasting — Albion Care Network

Week 1, Day 3 — Power Query cleaning of the six source tables

Purpose & Guiding Principle

This document records every cleaning and shaping step applied to the six source CSV files in Power Query (within Power BI Desktop), and the reasoning behind each. It serves as the Day 3 data-cleaning QA record.

The guiding principle throughout: a missing value does not always mean broken data. In these datasets most blanks are structurally correct — they are empty because the underlying event did not happen. Those were labelled or left as-is, not filled. Only genuine data gaps were repaired. Each column below is marked accordingly.

FIX = a real gap that was repaired. **LEAVE / LABEL** = blank by design; left as null or given an explicit label.

A global step applied to every table: all date and datetime columns were set using the English (United Kingdom) locale, so day/month values are not inverted (Power BI defaults to the US month/day order).

Summary of Changes

Table	Rows	Key cleaning action
`bed_inventory_occupancy`	701,760	Fill-down on missing occupied_beds
`admissions_discharges`	130,993	Derived LOS_hours; labelled destinations; deduped
`ed_outpatient_arrivals`	361,975	Left time_to_admission null (by design)
`elective_surgery_schedule`	25,355	Median-by-group fill; labelled cancellations
`staffing_resource`	116,960	No missing data; confirmed types only
`hospital_reference`	5	Lookup table; confirmed types only

1. bed_inventory_occupancy

The core fact table: occupied vs staffed beds by hospital, ward and bed type, at hourly grain across Jan 2024 – Dec 2025.

FIX — occupied_beds (2,105 nulls, 0.3%)

A small number of hours had no logged occupancy. These are a genuine recording gap, not empty wards. Because occupancy changes very little from one hour to the next, the previous hour's value was carried forward using Fill Down.

- **Action:** right-click occupied_beds column → Fill → Down.
- **Rationale:** filling with 0 would have falsely claimed empty wards and depressed the occupancy rate. Fill-down preserves the true operating level.

VERIFIED — no impossible values

A check confirmed zero rows where occupied_beds exceeds staffed_beds, so there were no overcapacity data errors to correct.

- **datetime:** confirmed as Date/Time type (UK locale).

2. admissions_discharges

One row per patient admission, including admission and discharge timestamps. Source of Length of Stay (LOS).

FIX — length_of_stay_hours (2,365 nulls, 1.8%) → recomputed

LOS was not truly missing — in every affected row both the admission and discharge timestamps were present, so LOS could be calculated exactly rather than estimated. A new column was derived and the original incomplete column removed.

- **Action:** Add Column → Custom Column, named LOS_hours, formula:

```
Duration.TotalHours([discharge_datetime] - [admission_datetime])
```

- **Then:** set LOS_hours to Decimal Number; removed the original length_of_stay_hours column.
- **Rationale:** computing from source timestamps is exact and preferable to imputation.

LABEL — discharge_destination (3,286 nulls, 2.5%)

Blank destinations were labelled rather than dropped, so they group cleanly in later analysis.

- **Action:** right-click column → Replace Values → replace blank with "Unknown".
- **Types:** admission_datetime and discharge_datetime set to Date/Time (UK locale) before deriving LOS.

3. ed_outpatient_arrivals

One row per Emergency Department / outpatient arrival — the emergency-inflow signal.

LEAVE — time_to_admission (255,459 nulls, 70.6%) — blank by design

This column is populated only for patients who were admitted. It is correctly blank for everyone discharged home, referred to outpatient, transferred, or who left without being seen — those patients were never admitted, so there is no time-to-admission to record. The nulls were left untouched.

- **Rationale:** filling with 0 or an average would fabricate admissions that never occurred. This column is only ever averaged over admitted patients.
- **arrival_datetime:** confirmed as Date/Time type (UK locale).

4. elective_surgery_schedule

One row per scheduled elective surgery, including expected length of stay and cancellation status.

FIX — expected_los_days (507 nulls, 2.0%) → median by procedure group

Expected LOS varies enormously by procedure — day-case procedures (Endoscopy, MRI Day Case) have an expected LOS of about 0.2 days, while major surgery (Tumour Resection) is around 4 days. A single global fill would therefore be wrong for most rows. Instead each blank was filled with the median expected LOS of its own procedure group.

This was done by building a small helper query and merging it back:

- **Step 1:** created LOS_medians — a helper query reading the same CSV, then Group By procedure_group with operation Median on expected_los_days (17 rows, one median per group).
- **Step 2:** merged LOS_medians back onto the main table on procedure_group (Left Outer join) and expanded the median_los column.
- **Step 3:** added a column filling blanks with the group median, keeping real values otherwise:

```
if [expected_los_days] = null then [median_los] else [expected_los_days]
```

- **Step 4:** removed helper columns and renamed the clean column back to expected_los_days.

Note on structure: LOS_medians was set to read the CSV directly (not to reference the main query). Referencing the main query while the main query merged the helper created a cyclic reference, which Power Query rejects. Reading the file independently breaks the loop.

- **Rationale for median over mean:** expected LOS is right-skewed (a few long stays up to 22.5 days), so the median reflects the typical case without being pulled up by outliers.

LABEL — cancellation_reason (23,365 nulls, 92.2%)

Blank for every completed surgery (no cancellation, so no reason) and populated for every cancelled one — a perfectly consistent field, not a data gap. Blanks were labelled for clarity.

- **Action:** Replace Values → blank replaced with "Not Cancelled".
- **surgery_date:** confirmed as Date type (ISO format, parsed cleanly).

5. staffing_resource

Daily planned vs actual staffing by hospital, ward and role, with a safe-ratio flag.

NO CHANGES — zero missing values

- **date:** confirmed as Date type (UK locale).
- **Other columns:** complete and correctly typed; planned_staff and actual_staff whole numbers, safe_ratio_met as Yes/No text. No cleaning required.

6. hospital_reference

The lookup / dimension table describing each of the five hospitals (name, city, tier, specialty emphasis, capacity).

NO CHANGES — zero missing values

- **Confirmed:** total_bed_capacity as whole number; text columns as text. Left as the clean dimension table for the model.

7. Duplicate Check (all tables)

Every table was checked for duplicate records — both fully-identical rows and repeated key values (e.g. repeated admission_id, or repeated hospital/ward/bed-type/hour combinations).

CLEAN — five of six tables

bed_inventory_occupancy, ed_outpatient_arrivals, elective_surgery_schedule, staffing_resource and hospital_reference contained no duplicate rows. In particular, bed_inventory_occupancy had no repeated hospital/ward/bed-type/hour combination, confirming one clean reading per hour.

FIX — admissions_discharges (486 identical duplicates removed)

This table contained 486 fully-identical duplicate rows (entire row repeated, including admission_id). These would double-count admissions, so they were removed.

- **Action:** selected all columns → Home → Remove Rows → Remove Duplicates. Row count fell from 131,479 to 130,993 (−486).

RETAINED — 37 conflicting-key rows

A further 37 rows shared an admission_id but differed in other fields (for example a different ward or discharge time) — conflicting records rather than clean copies. Because it is ambiguous which version is correct, these were retained as distinct records rather than arbitrarily dropped, and flagged as a minor data-quality note (0.03%).

- **Rationale:** removing exact duplicates is safe and necessary; force-deduplicating conflicting records would mean guessing which row is valid, so they were kept and documented instead.

Loading into the Model

After the steps above, all six queries were loaded into the data model via Home → Close & Apply. Every cleaned column was verified in the Column Profile pane to show the expected result (for example, occupied_beds, LOS_hours and expected_los_days all showing 0% empty after their fixes). This completes the "Clean analysis tables" output for Day 3.

Note on data scope: the reference table contains five hospitals; the wider Albion Care Network is larger. This project analyses the provided five-hospital sample, stated as a limitation in the final report.